

Linear Transformations

Geometric Algorithms
Lecture 7

Practice Problem

Find three vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^3$ such that

- » every pair of vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$, $\{\mathbf{v}_1, \mathbf{v}_3\}$, $\{\mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent
- » $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent

$$x\vec{v}_1 + y\vec{v}_2 = \vec{0}$$

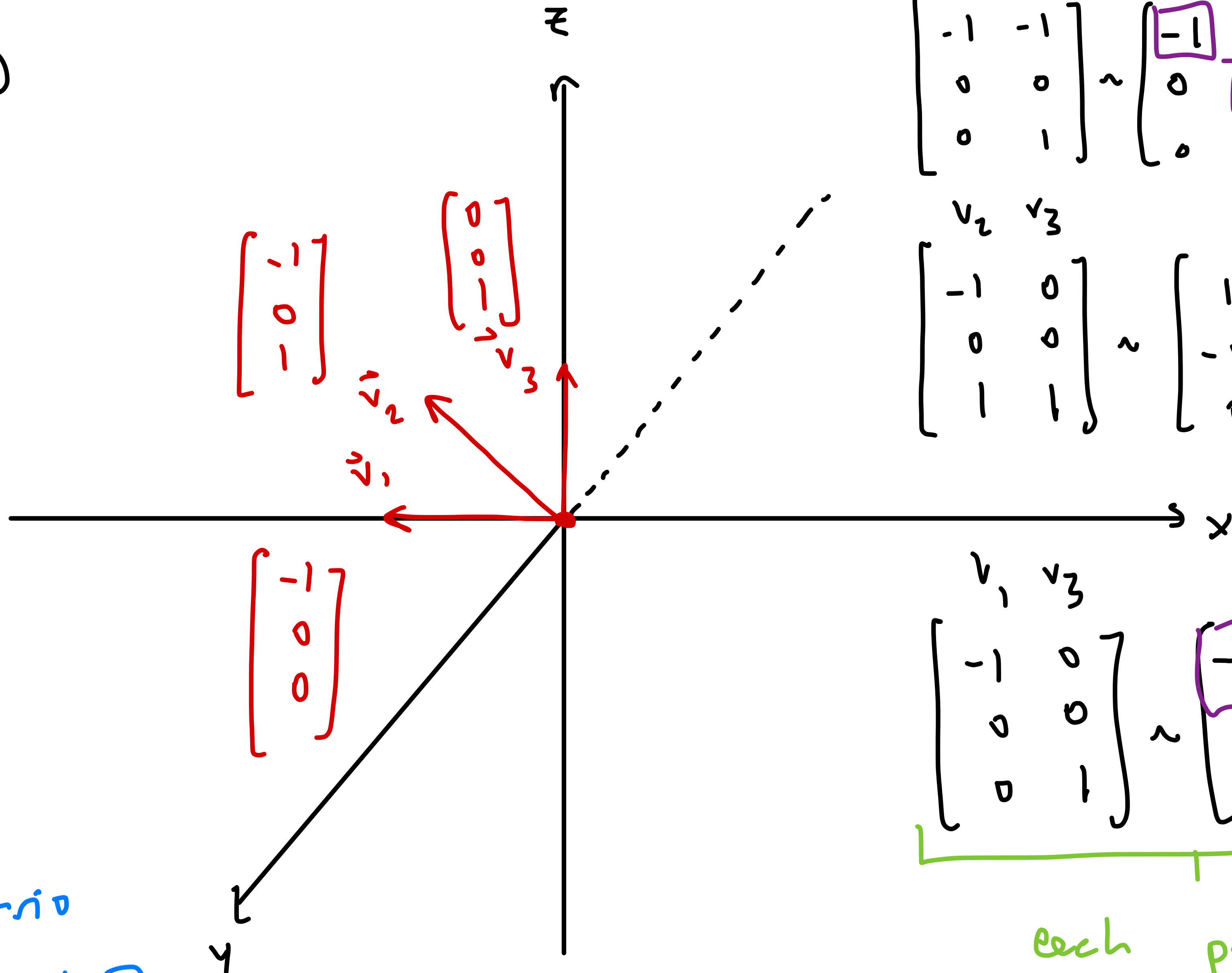
$$x = 0$$

$$y = 5000$$

$$\begin{bmatrix} -1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \sim$$

$$\begin{bmatrix} \boxed{-1} & -1 & \boxed{0} \\ 0 & \boxed{1} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

trio
is L.I.



$$\begin{bmatrix} v_1 & v_2 \\ -1 & -1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \sim \begin{bmatrix} \boxed{-1} & -1 \\ 0 & \boxed{1} \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} v_2 & v_3 \\ -1 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & 1 \\ 0 & \boxed{1} \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} v_1 & v_3 \\ -1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \sim \begin{bmatrix} \boxed{-1} & 0 \\ 0 & \boxed{1} \\ 0 & 0 \end{bmatrix}$$

each pair is
L.I.

Objectives

- » Introduce matrix transformations
- » Define linear transformations
- ~~» Start looking at the geometry of linear transformations~~

Keywords

Transformations

Domain, Codomain

Image, Range

Matrix Transformations

Linear Transformations

Additivity, Homogeneity

Dilation, Contraction, Shearing, Rotation

Matrix Transformations

Recall: Spans (with Matrices)

$$\text{span} \{ \vec{v}_1, \dots, \vec{v}_n \} = \{ \overbrace{\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n}^{\text{lin comb.}} : \alpha_1, \dots, \alpha_n \in \mathbb{R} \}$$

$$V = \begin{bmatrix} | & & | \\ \vec{v}_1 & \dots & \vec{v}_n \\ | & & | \end{bmatrix} \quad V \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 \vec{v}_1 + \dots + x_n \vec{v}_n$$

$$= \{ V \vec{x} : \vec{x} \in \mathbb{R}^n \}$$

Recall: Spans (with Matrices)

Df. The *span* of a set of vectors is the set of all possible linear combinations

$$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \{[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{v} : \mathbf{v} \in \mathbb{R}^n\}$$

Matrices as Transformations

$$\vec{x} \mapsto A \vec{x}$$

$$T(\vec{x}) = \underbrace{A}_{m \times n} \underbrace{\vec{x}}_{\mathbb{R}^n}$$

$$m \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

Matrices as Transformations

Matrices *transform* vectors into vectors the span of its columns

$$\mathbf{x} \mapsto A\mathbf{x}$$

map a vector $\mathbf{v}$ to the vector $A\mathbf{v}$

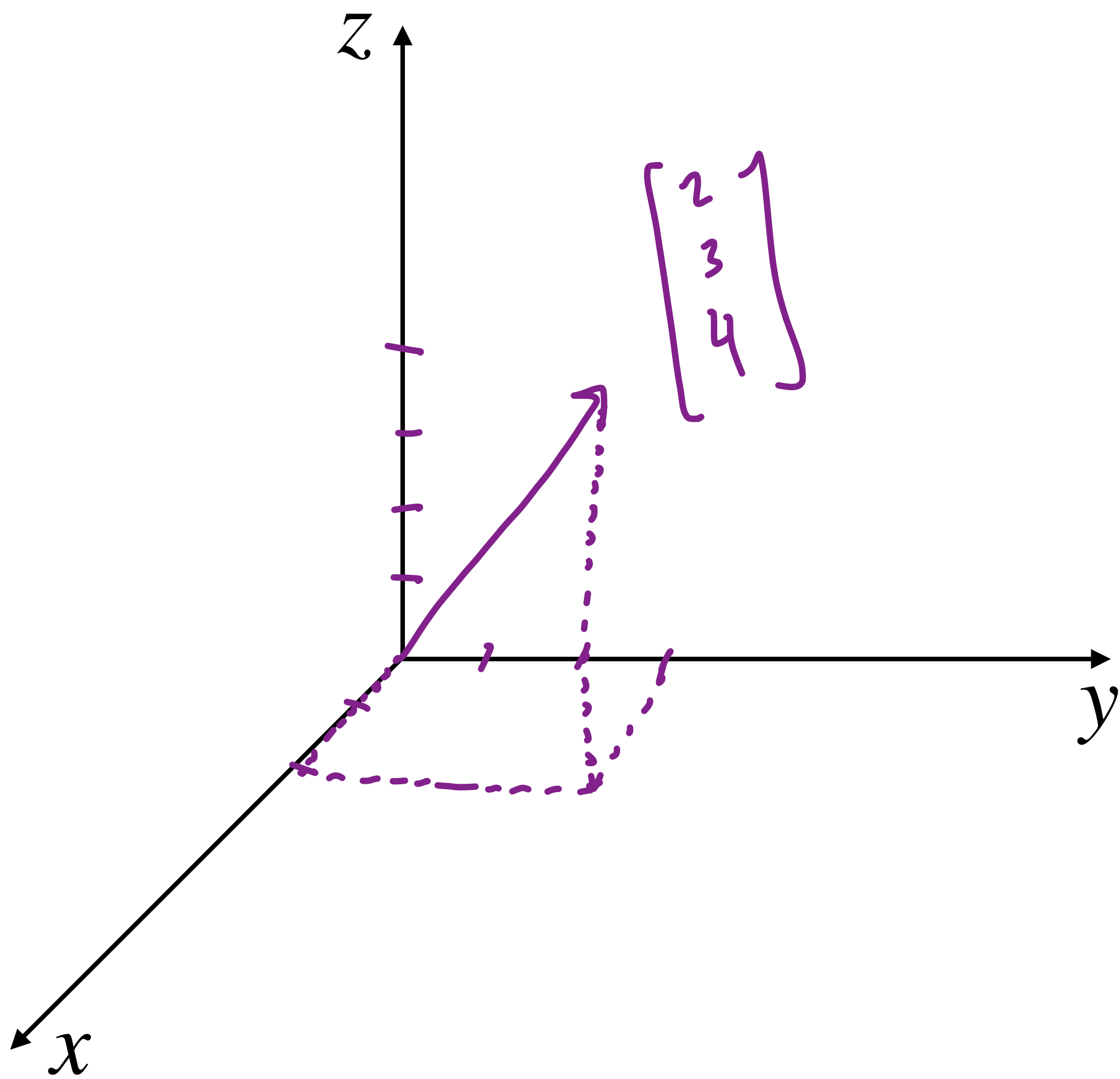
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

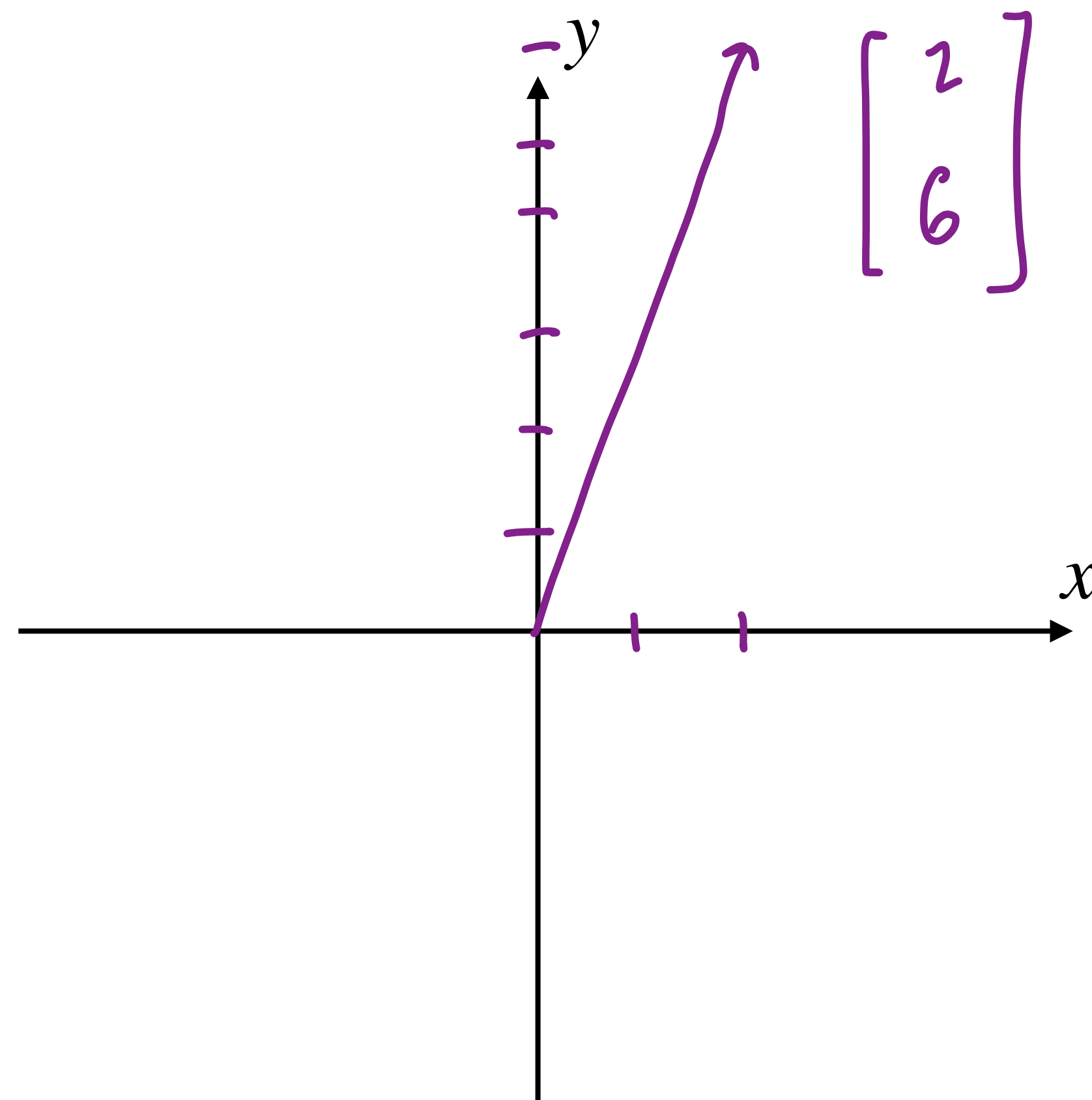
$$= \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \mathbf{x}$$

A blue arrow points from the matrix equation to the right, indicating a transformation or mapping.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}$$



Important

The vector may be a different
size after translation

Recall: Matrix-Vector Multiplication

matrix-vector multiplication only works if the number of *columns* of the matrix matches the dimension of the vector

$$\begin{array}{c} \textcolor{blue}{m} \left[\begin{array}{ccc} * & \dots & * \\ * & \dots & * \\ \vdots & \ddots & \vdots \\ * & \dots & * \\ * & \dots & * \end{array} \right] \quad \textcolor{red}{n} \left[\begin{array}{c} * \\ \vdots \\ * \end{array} \right] = \textcolor{blue}{m} \left[\begin{array}{c} * \\ * \\ \vdots \\ * \\ * \end{array} \right] \end{array}$$

$(\textcolor{blue}{m} \times \textcolor{red}{n}) \quad \mathbb{R}^{\textcolor{red}{n}} \quad \mathbb{R}^{\textcolor{blue}{m}}$

Motivating Questions

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New Interpretation

$$\vec{x} \mapsto A \vec{x}$$

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A transforms into $\mathbf{b}$?

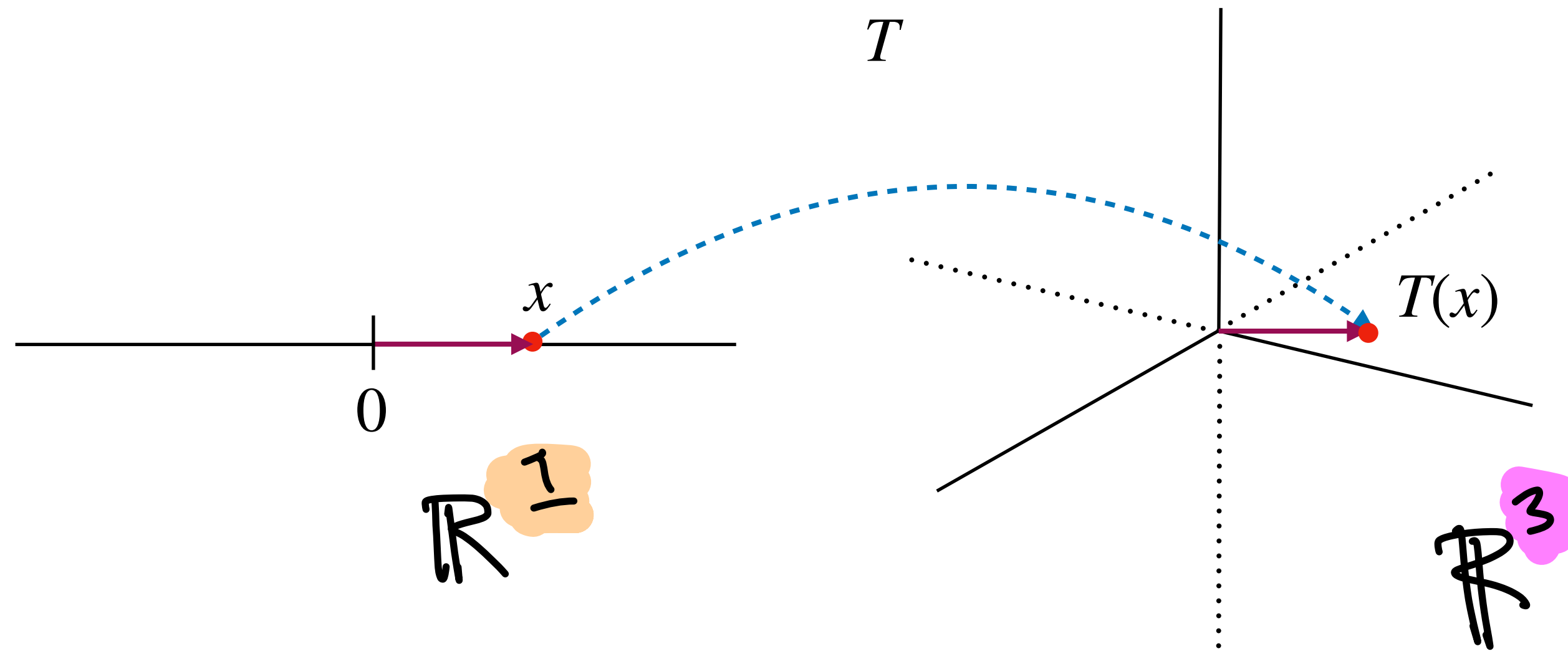
Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A transforms into $\mathbf{b}$

Question

Suppose a matrix transforms vectors according to the following picture. *What's the shape of the matrix?*

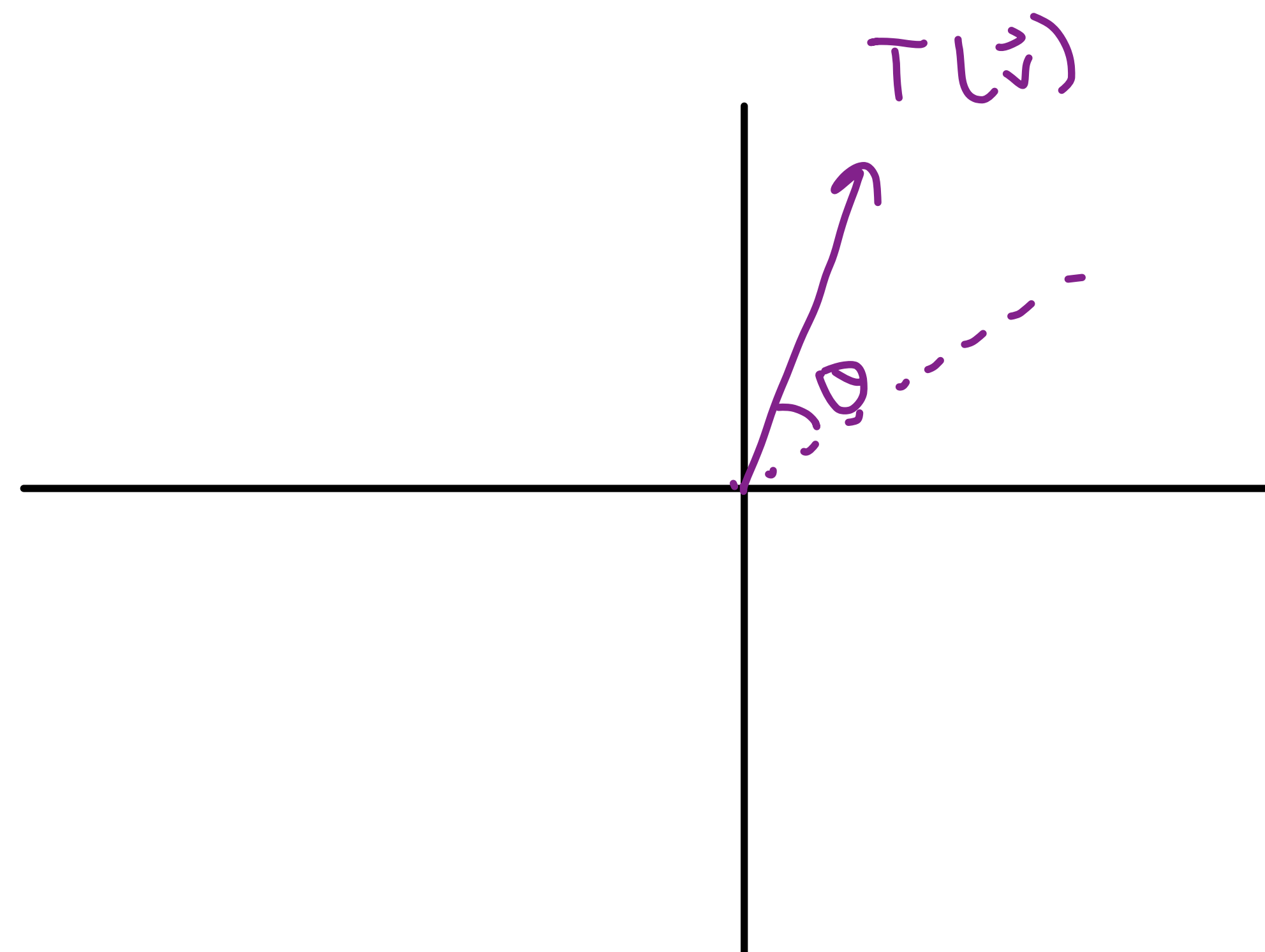
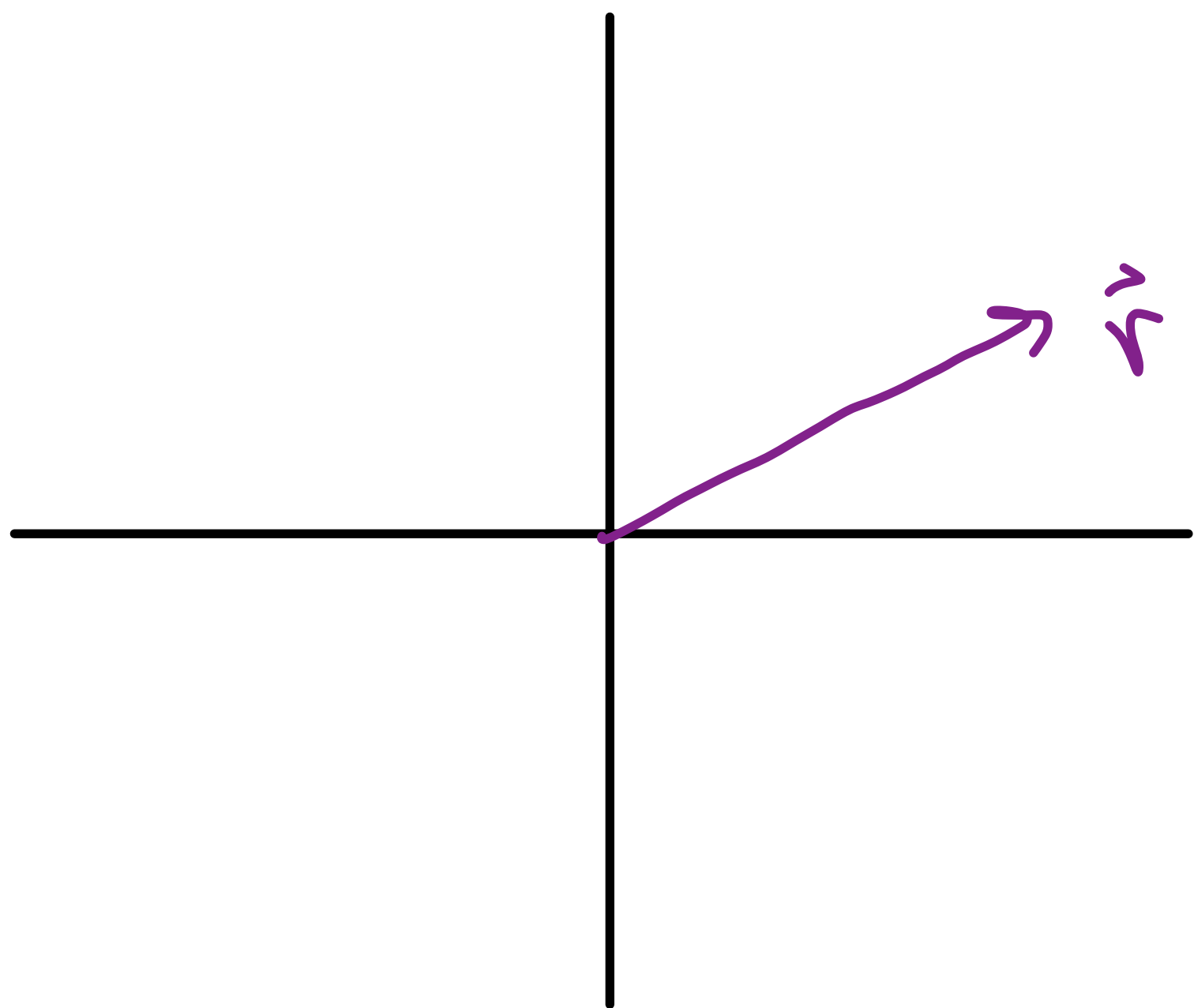
$$A = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$T(\vec{x}) = A \vec{x}$$



$$\mathbb{R}^n \rightarrow \mathbb{R}^n$$

Mapping between the same space can be viewed as a way of "moving around points"



Transformations

Transformations in General

Df. A *transformation* T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a function which maps every vector $\mathbf{v}$ in $\mathbb{R}^n$ to a vector $T(\mathbf{v})$ in $\mathbb{R}^m$

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$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

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$$\begin{array}{ccc} T : \mathbb{R}^n & \longrightarrow & \mathbb{R}^m \\ \text{domain} & & \text{codomain} \\ \text{source space} & & \text{target space} \\ \text{dom}(T) & & \text{cod}(T) \end{array}$$

Image and Range

Df. The image of $\vec{v}$ under T is $T(\vec{v})$, $\text{im}_T(\vec{v})$

ex: $T(\vec{x}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \vec{x}$ $\text{im}_T\left(\begin{bmatrix} 2 \\ 3 \\ 12 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$

Def. The range of $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is $\{T(\vec{v}) : v \in \mathbb{R}^n\}$

ex: $\text{ran}(T) = \mathbb{R}^2$ $\begin{bmatrix} 7 \\ 13 \end{bmatrix} \in \text{ran}(T)$ since

$$T\left(\begin{bmatrix} 7 \\ 13/2 \\ \pi \end{bmatrix}\right) = \begin{bmatrix} 7 \\ 13 \end{bmatrix}$$

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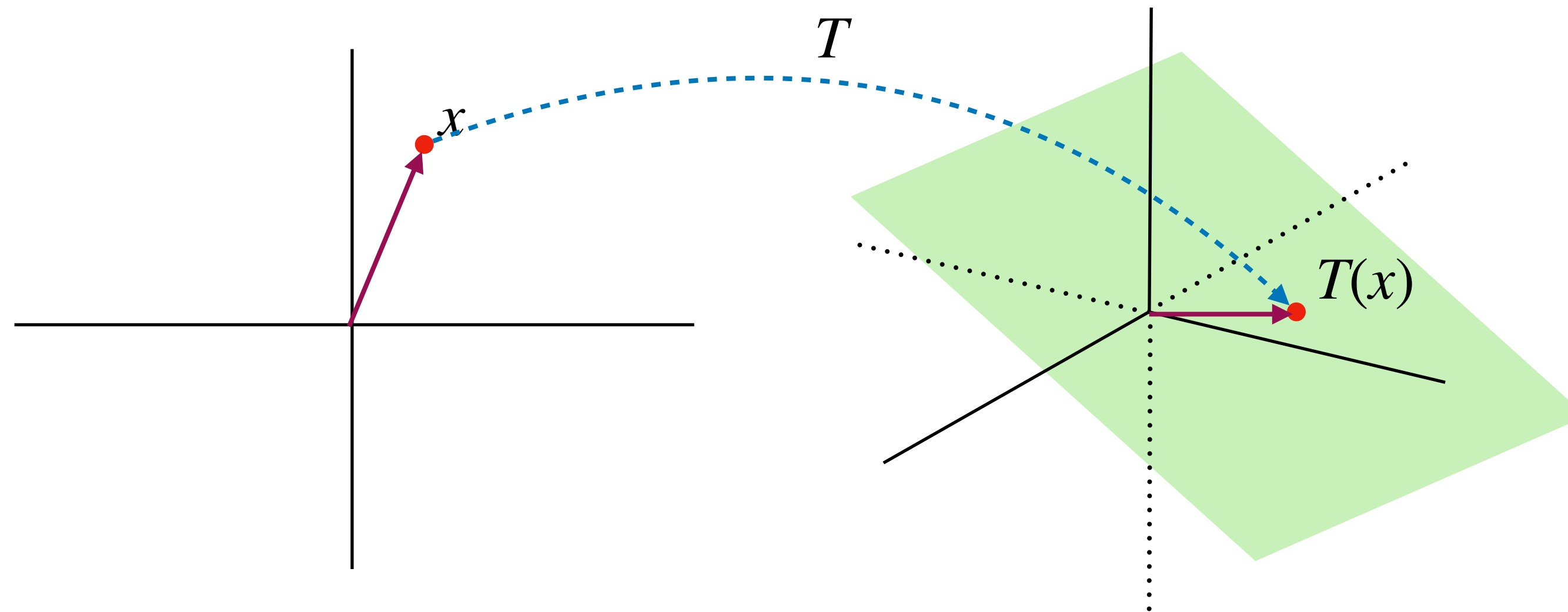
$$\text{ran}(T) = \{T(\mathbf{v}) : \mathbf{v} \in \mathbb{R}^n\}$$

image of $\mathbf{v}$ under $T \equiv$ output of T applied to $\mathbf{v}$

range of $T \equiv$ all possible output of T

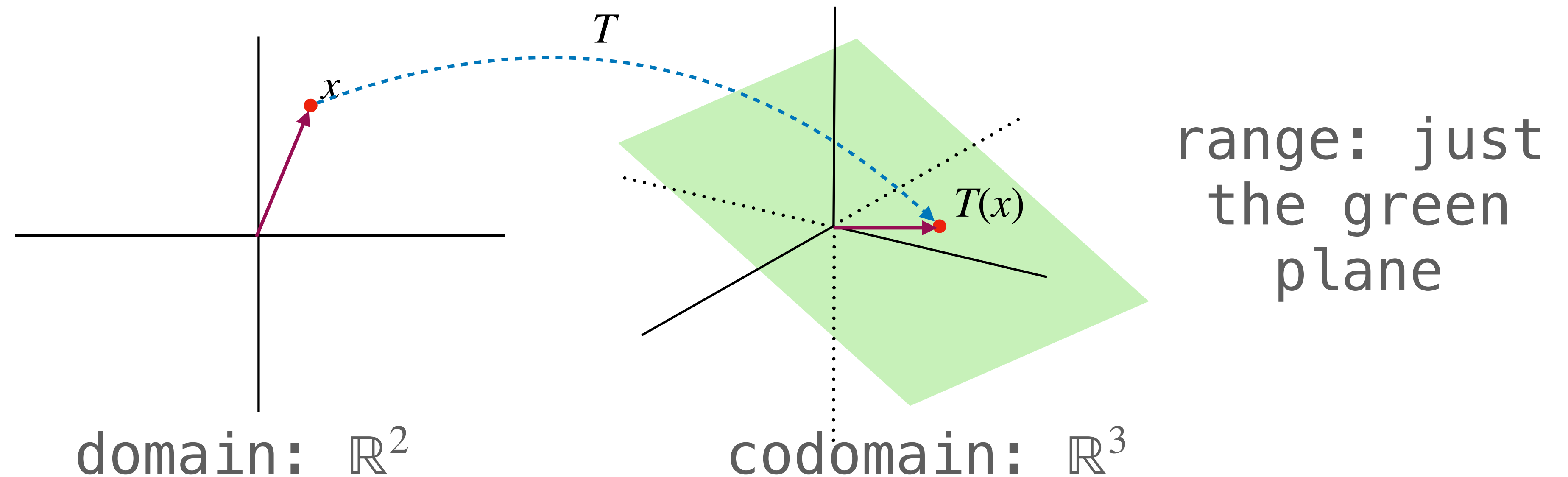
Codomain and Range

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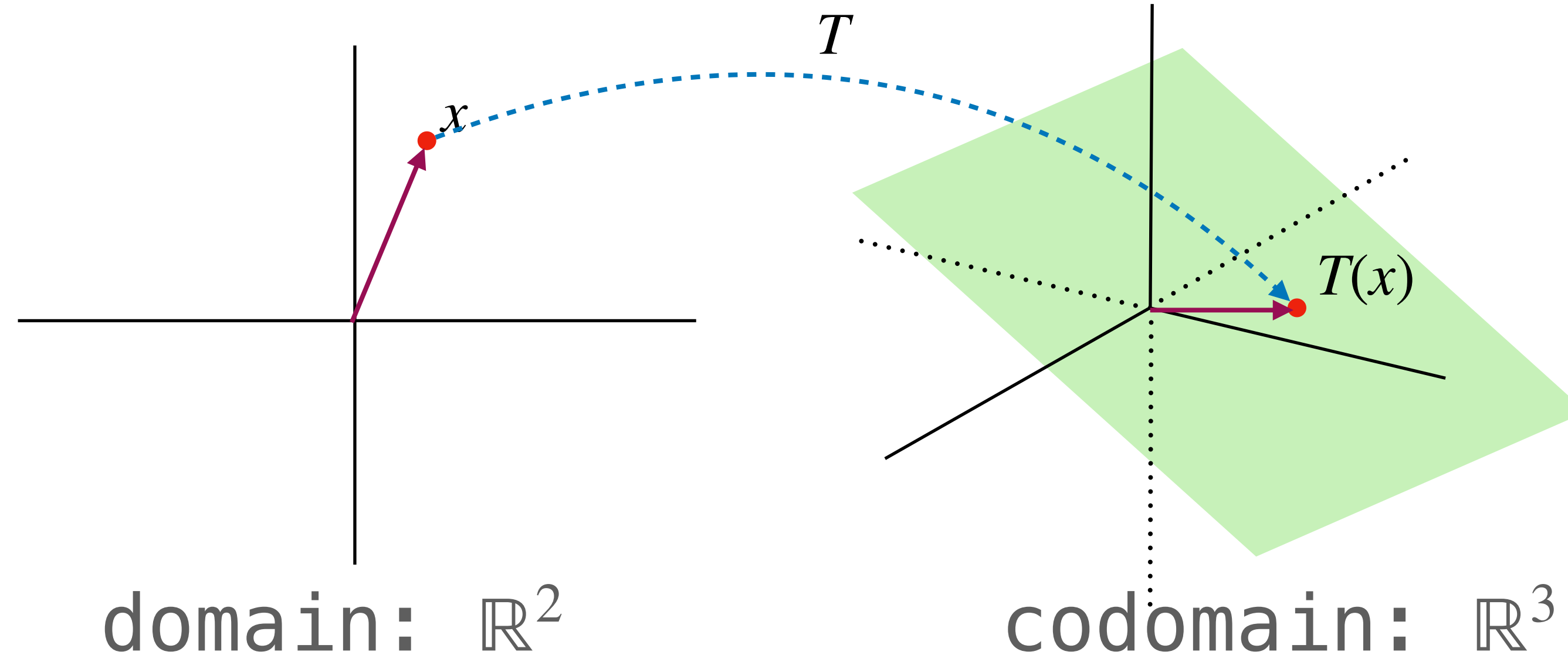


Codomain and Range

The codomain and range of a transformation may
or may not be the same

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \notin \text{ran}(T)$$



range: just
the green
plane

The range is always contained in the codomain

Example

T

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1^2 \\ x_2 \\ 0 \end{bmatrix} \quad \begin{bmatrix} \sqrt{11} \\ -15 \end{bmatrix} \mapsto \begin{bmatrix} 11 \\ -15 \\ 0 \end{bmatrix}$$

domain : $\mathbb{R}^2$

codomain : $\mathbb{R}^3$

range :

$$\begin{aligned} \text{ran}(T) &= \left\{ \begin{bmatrix} x_1^2 \\ x_2 \\ 0 \end{bmatrix} : x_1, x_2 \in \mathbb{R} \right\} \\ &= \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x \geq 0 \right\} \end{aligned}$$

Matrix Transformations

Transformation of a Matrix

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Df. The *transformation of a* $(m \times n)$ *matrix* A is the function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$T(\mathbf{v}) = A\mathbf{v}$$

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i.e., Given $\mathbf{v}$, return A multiplied by $\mathbf{v}$

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i.e., Given $\mathbf{v}$, return A multiplied by $\mathbf{v}$

e.g., $T(\mathbf{v}) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{v}$

Range and Span

$$\underbrace{\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}} = \underbrace{\text{ran}([\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n])}$$

$$\{x_1 \vec{a}_1 + \dots + x_n \vec{a}_n : x_1, \dots, x_n \in \mathbb{R}\} \quad \{[\vec{a}_1, \dots, \vec{a}_n] \vec{x} : x \in \mathbb{R}^n\}$$

equal

Linear Transformations

Motivating Questions

What kind of functions can we define this way?

How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

Recall: Algebraic Properties of Matrix-Vector Multiplication

(Additivity) $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$

(Homogeneity) $A(c\vec{u}) = c(A\vec{u})$

Recall: Algebraic Properties

Matrix-vector multiplication satisfies the following two properties:

1. $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ (additivity)

2. $A(c\mathbf{v}) = c(A\mathbf{v})$ (homogeneity)

Example

Linear Transformations

Df. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Linear Transformations

Df. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

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Matrix transformations are linear transformations

Example: Identity

$$T(\mathbf{v}) = \mathbf{v}$$

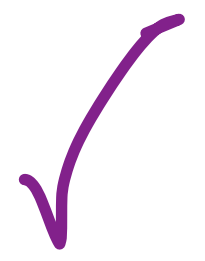
Linear



add

$$T(\mathbf{v} + \mathbf{u}) = \mathbf{v} + \mathbf{u} = \mathbf{v} + \mathbf{u} = T(\mathbf{v}) + T(\mathbf{u})$$

homog.

$$T(c\vec{v}) = c\vec{v} = c\vec{v} = cT(\vec{v})$$
A purple checkmark symbol indicating that the identity transformation is homogeneous.

Example: Zero

$$T(\mathbf{v}) = \mathbf{0}$$

Linear

$$T(\vec{u} + \vec{v}) = \vec{0} = \vec{0} = \vec{0} + \vec{0} = T(\vec{u}) + T(\vec{v}) \quad \checkmark$$

$$T(c\vec{v}) = \vec{0} = c\vec{0} = cT(\vec{v}) \quad \checkmark$$

not linear

$$T(\vec{v}) = \vec{i}$$

$$T(\vec{i} + \vec{i}) = \vec{i} \neq \vec{i} + \vec{i} = T(\vec{i}) + T(\vec{i})$$

Example

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 + x_2 \\ x_2 - x_3 \\ 3x_1 \end{bmatrix}$$

$$T \left(c \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = T \left(\begin{bmatrix} cx_1 \\ cx_2 \\ cx_3 \end{bmatrix} \right) = \begin{bmatrix} cx_1 + cx_2 \\ cx_2 - cx_3 \\ 3cx_3 \end{bmatrix}$$

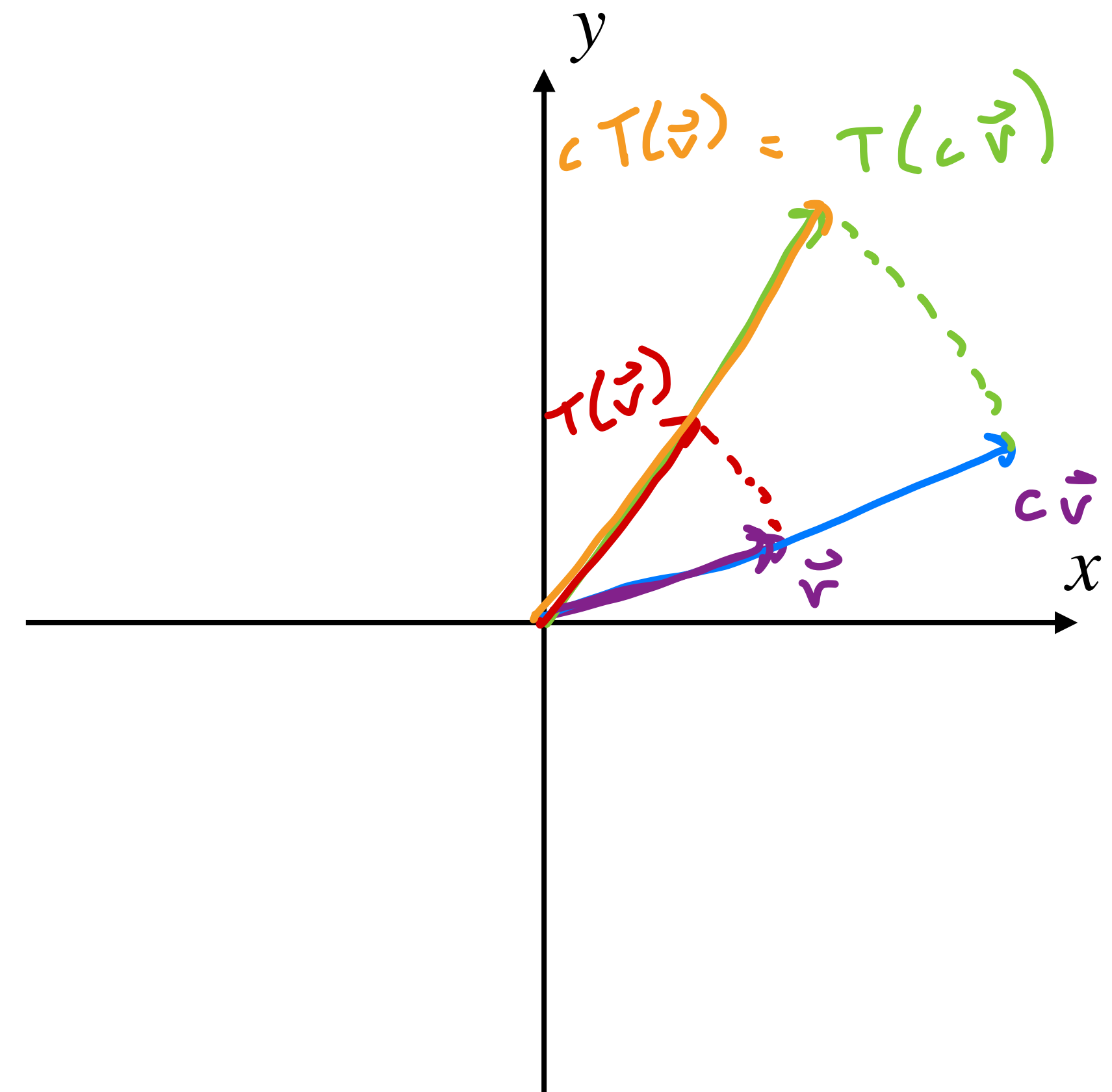
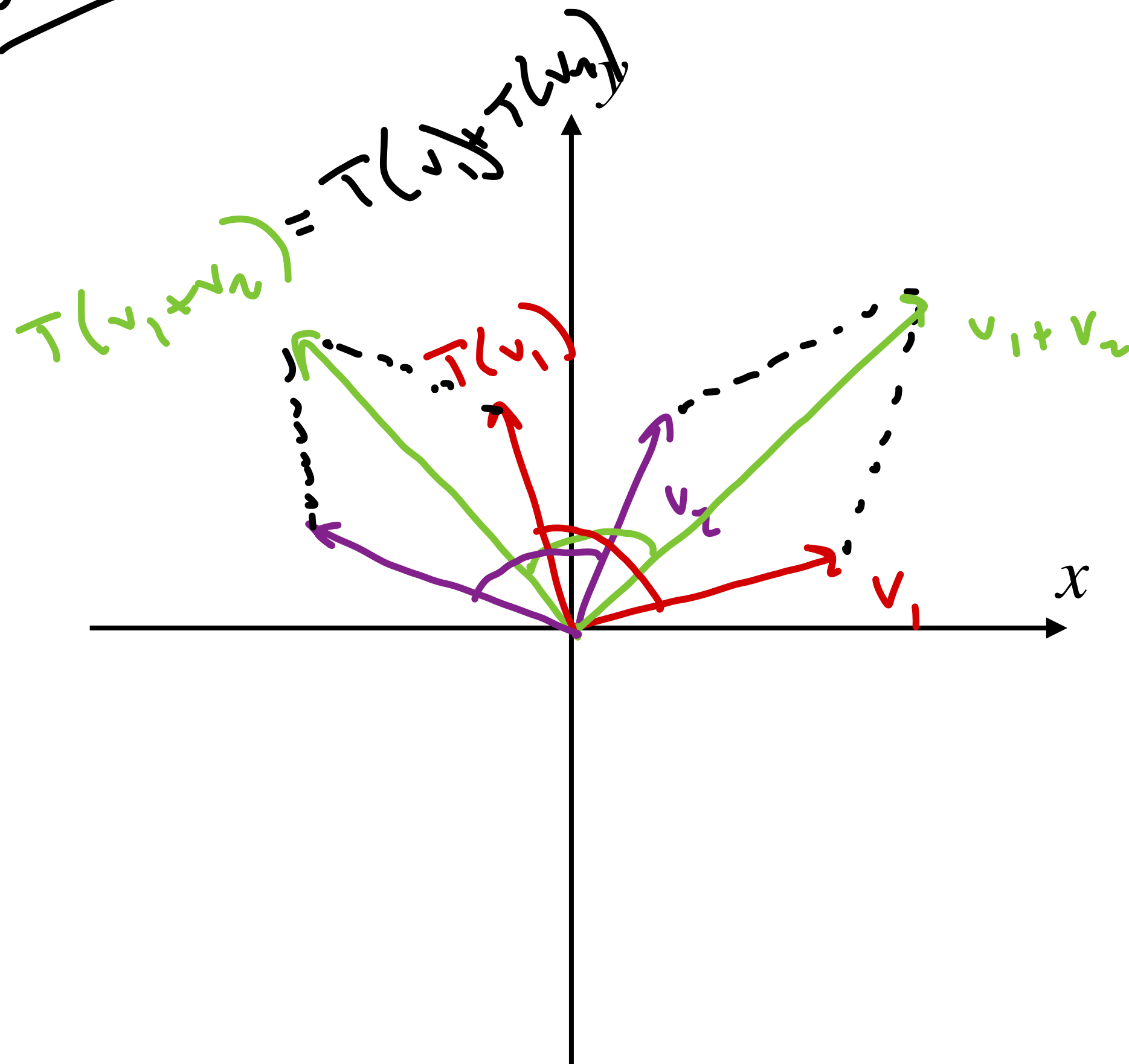
$$= \begin{bmatrix} c(x_1 + x_2) \\ c(x_2 - x_3) \\ c(3x_3) \end{bmatrix} = c T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right)$$

Example: Rotation

We'll see this next time, but for now we can use pictures

add.

homom.



Example: Indefinite Integrals

$$T(f) = \int f(x) dx$$

Disclaimer:
Advanced
Material

Example: Expectation

$$T(X) = \mathbb{E}[X]$$

Disclaimer:
Advanced
Material

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

Non-Example: Squares

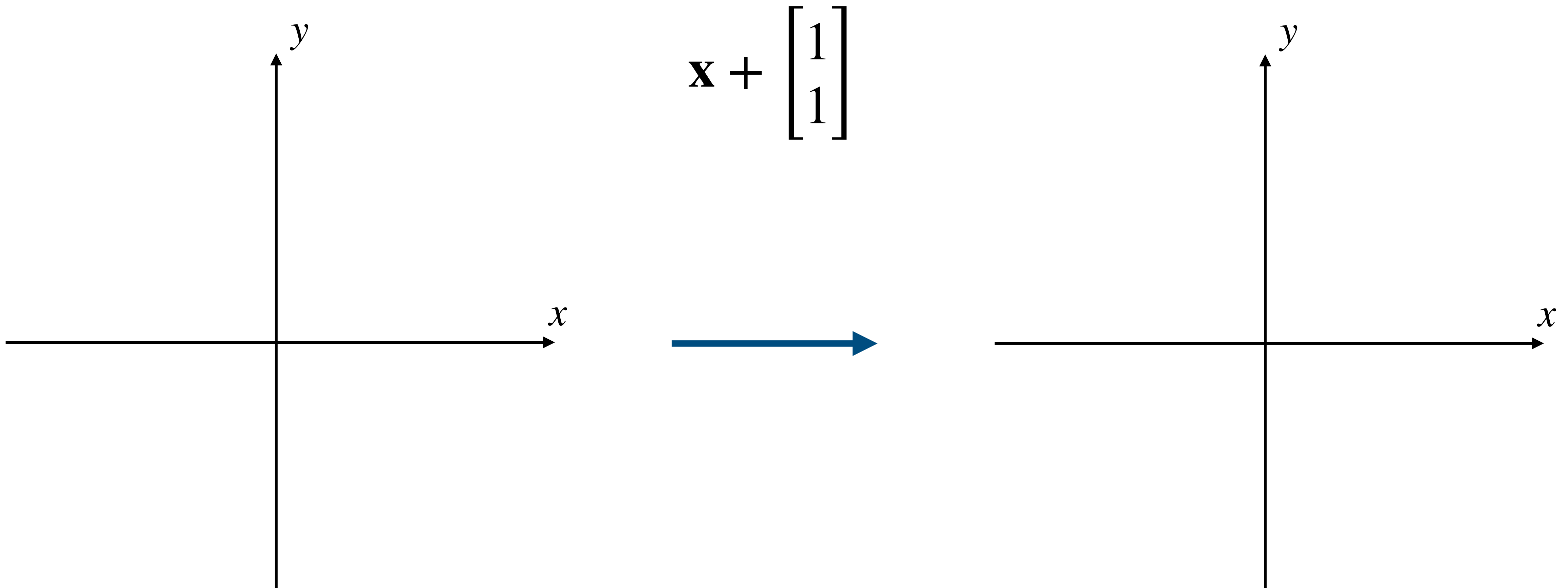
$$T(x) = x^2$$

$$T(2(3)) \quad T: \mathbb{R} \rightarrow \mathbb{R}$$

$$T(6) = 36$$

$$2T(3) = 18$$

Non-Example: Translation



Properties of Linear Transformations

The Zero Vector

The Zero Vector

$$T(\mathbf{0}) = ???$$

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$

$$T(\vec{0}) = T(0\vec{1}) = 0 + (\vec{1}) = \vec{0}$$

The zero vector is *fixed* by linear transformations

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$

Note: These may be different dimensions!

The zero vector is *fixed* by linear transformations

A Single Condition

Thm. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear if

for any $\vec{u}, \vec{v} \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$

$$T(a\vec{u} + b\vec{v}) = aT(\vec{u}) + bT(\vec{v})$$

Proof. exercise.

A Single Condition

$$T(a\mathbf{v} + b\mathbf{u}) = aT(\mathbf{v}) + bT(\mathbf{u})$$

A Single Condition

$$T(a\mathbf{v} + b\mathbf{u}) = aT(\mathbf{v}) + bT(\mathbf{u})$$

We can combine our linearity conditions

A Single Condition

Thm. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is linear if and only if for any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^m$ and any real numbers a and b ,

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$$

A Single Condition

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$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$$

It's often easiest to show this
single condition

Linear Combinations

$$T\left(a_1\vec{v}_1 + \dots + a_n\vec{v}_n\right) = a_1T(\vec{v}_1) + \dots + a_nT(\vec{v}_n)$$

$$T\left(\sum_{i=1}^n a_i\vec{v}_i\right) = \sum_{i=1}^n a_iT(\vec{v}_i)$$

Geometry of Matrix Transformations

Motivating Questions

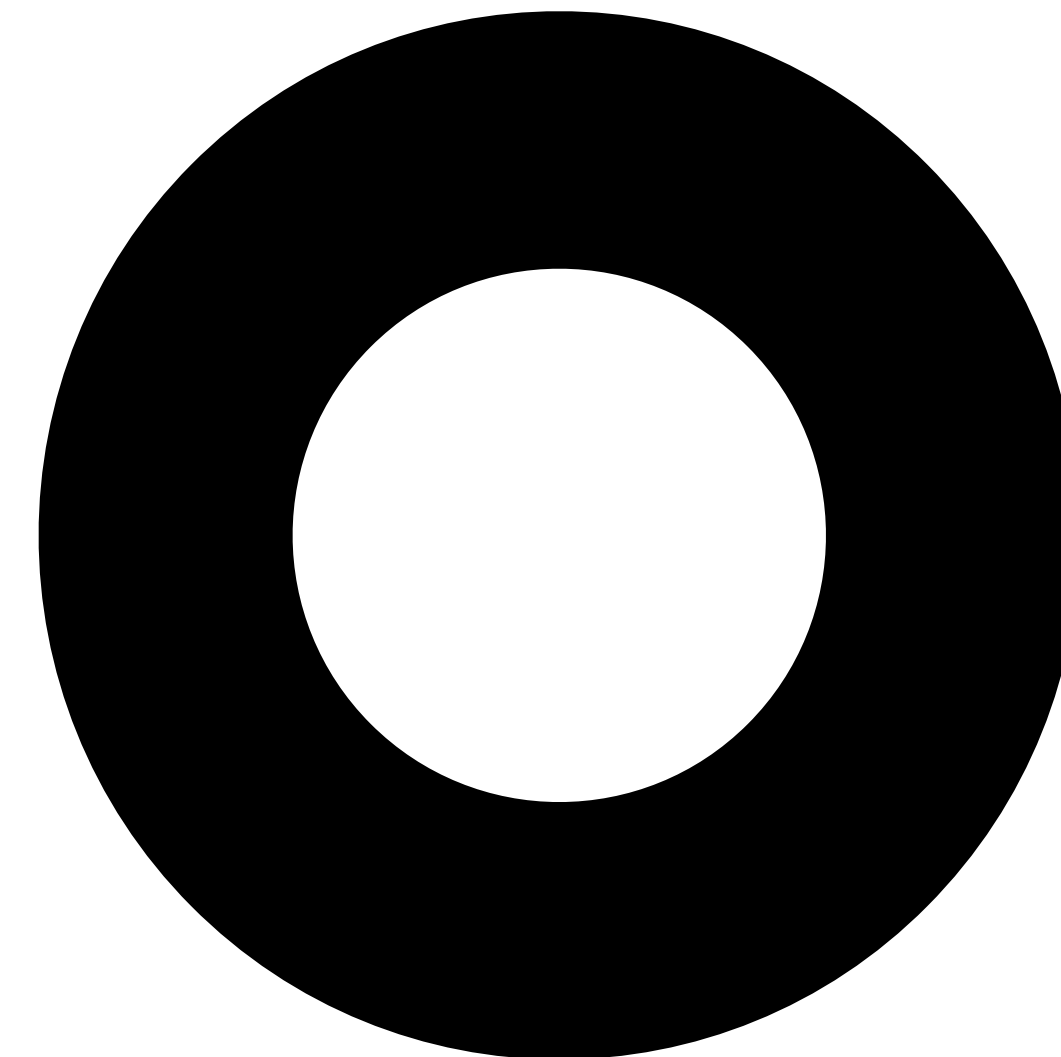
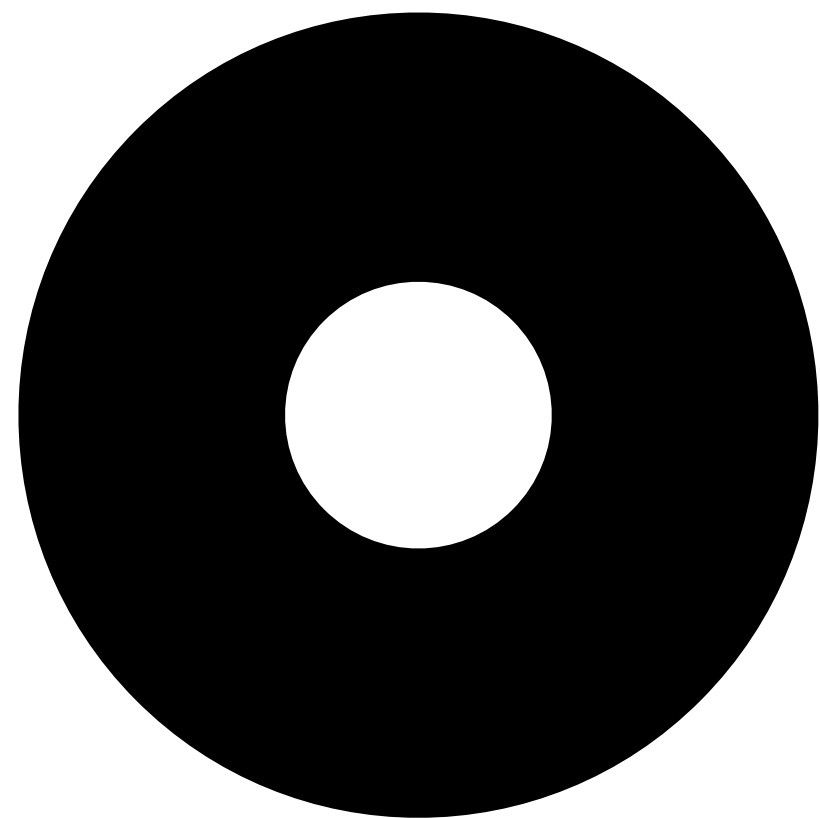
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How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

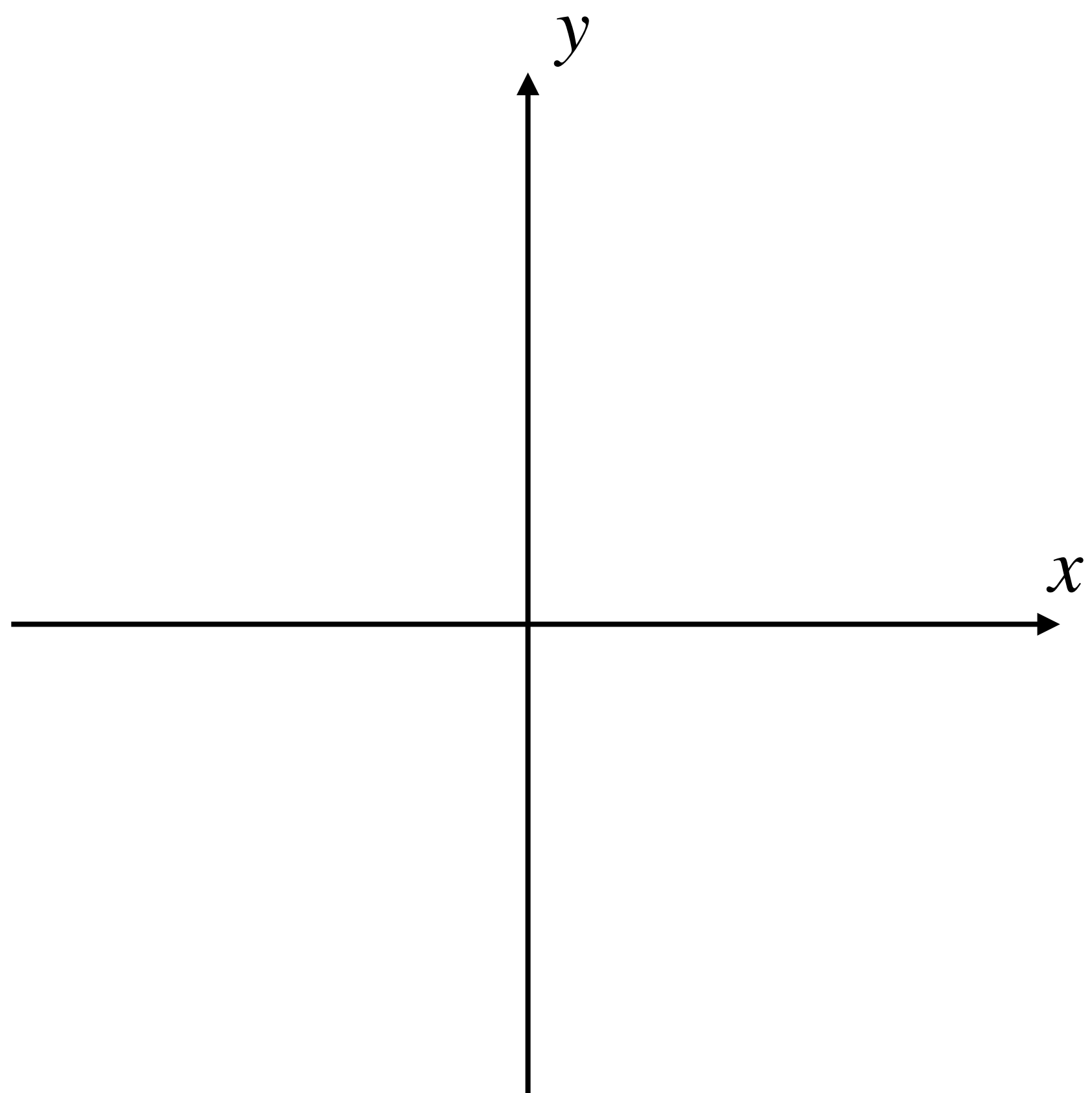
Matrix transformations change the
"shape" of a set of set of
vectors (points)

Example: Dilation

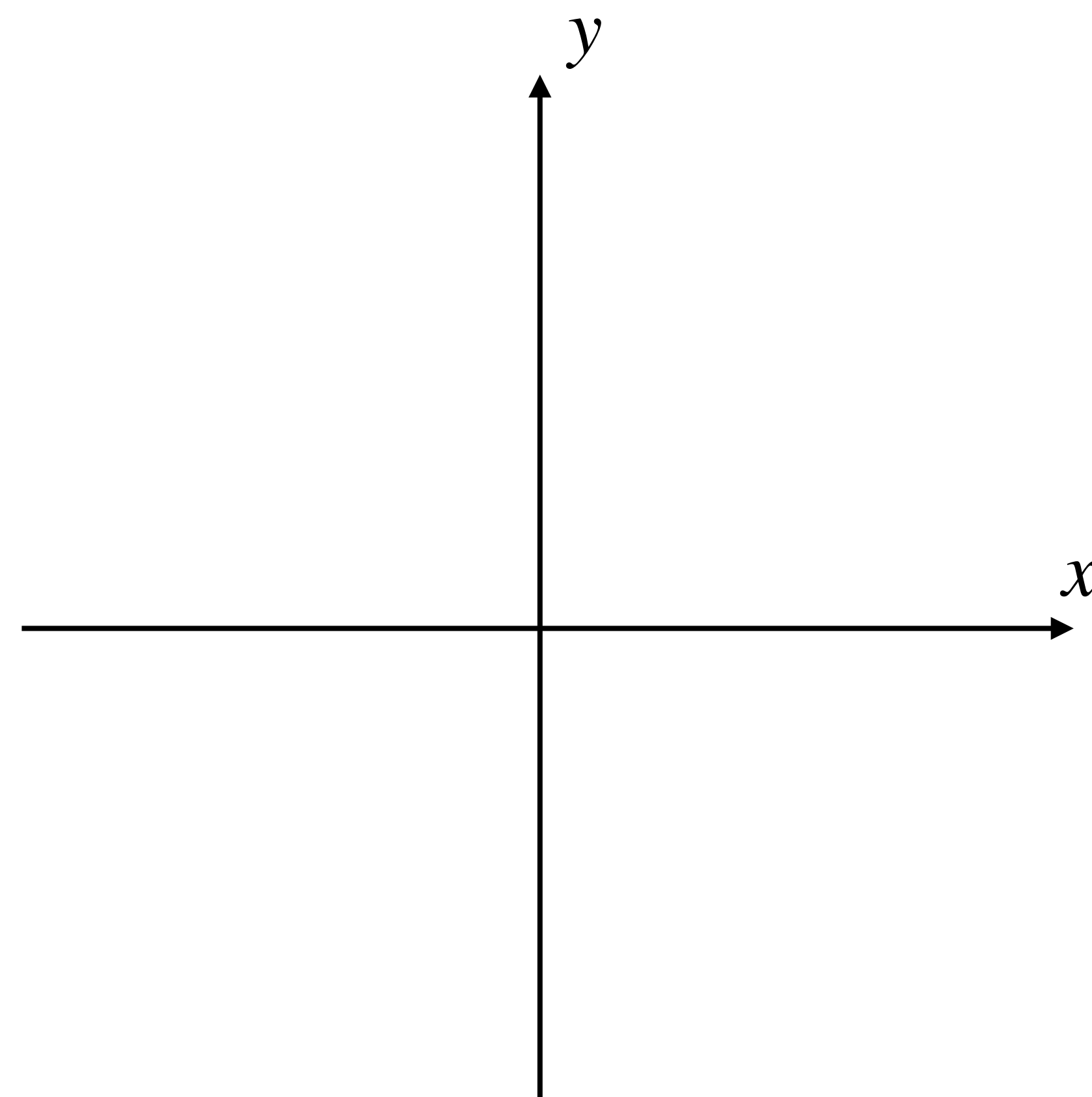


Example: Dilation

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$

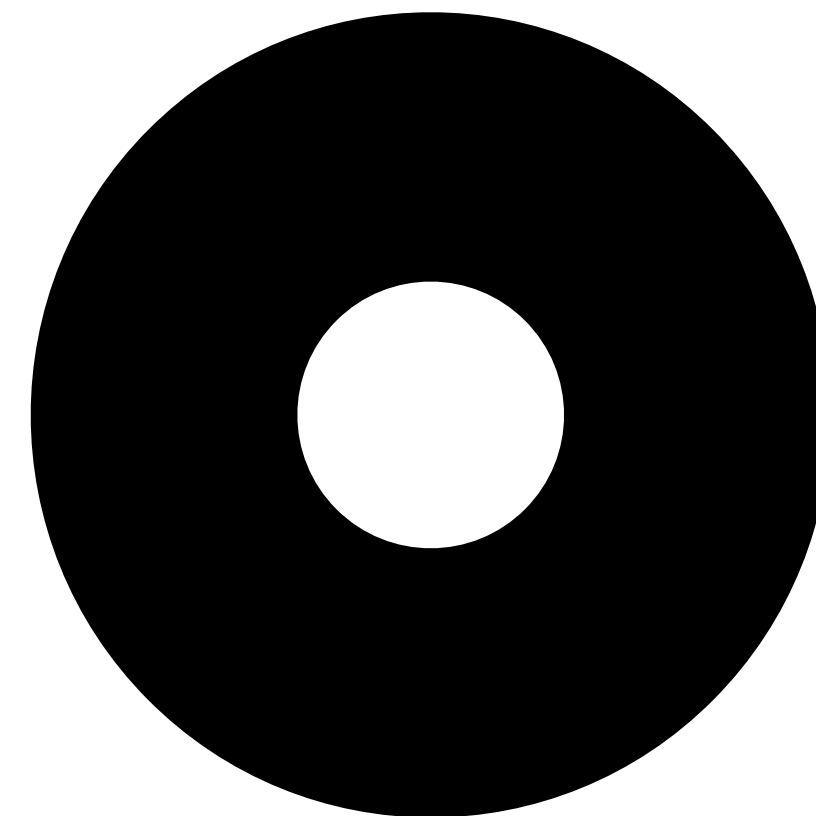
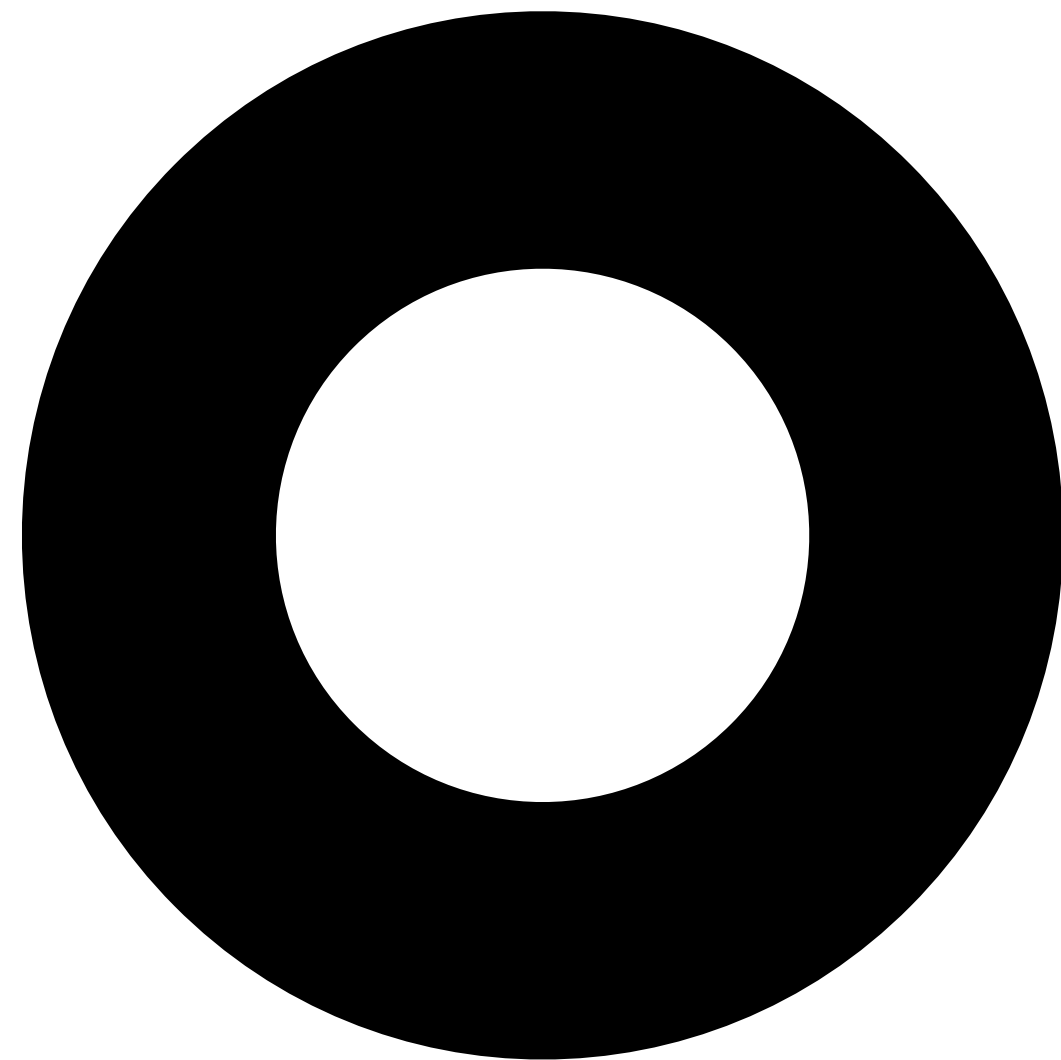


$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x}$$

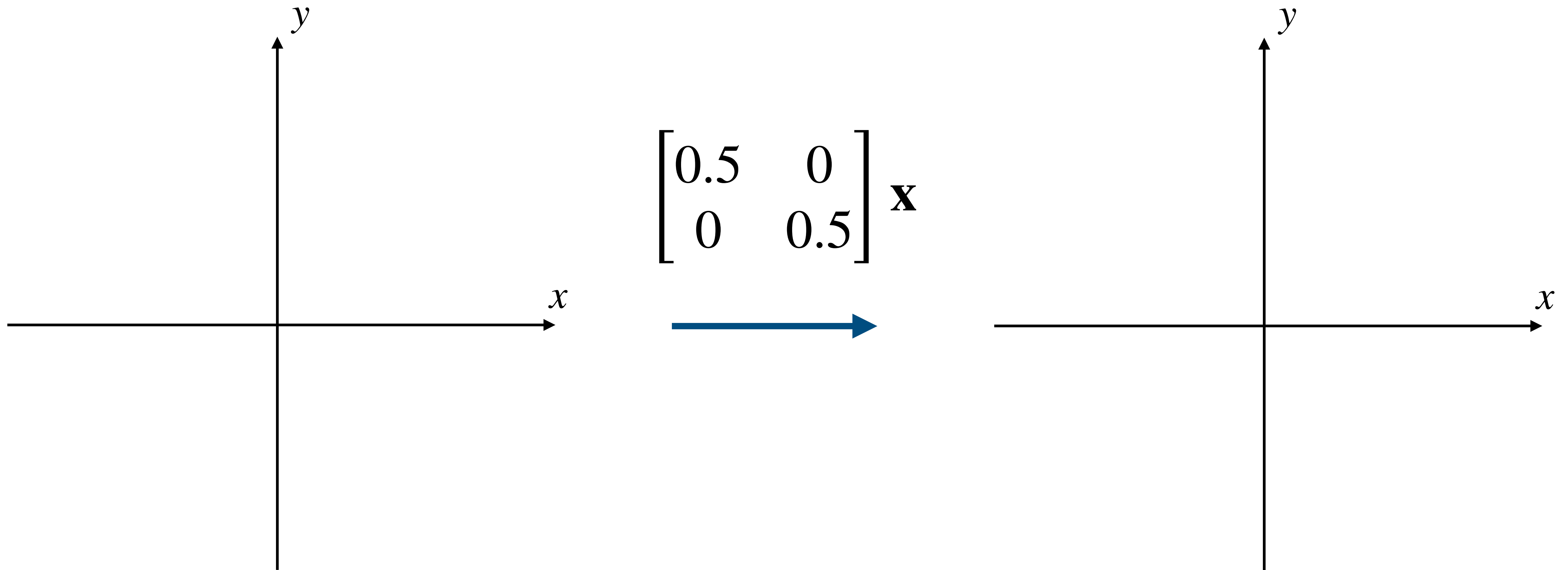
if $r > 1$, then the transformation pushes points away from the origin

Example: Contraction



Example: Contraction

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$



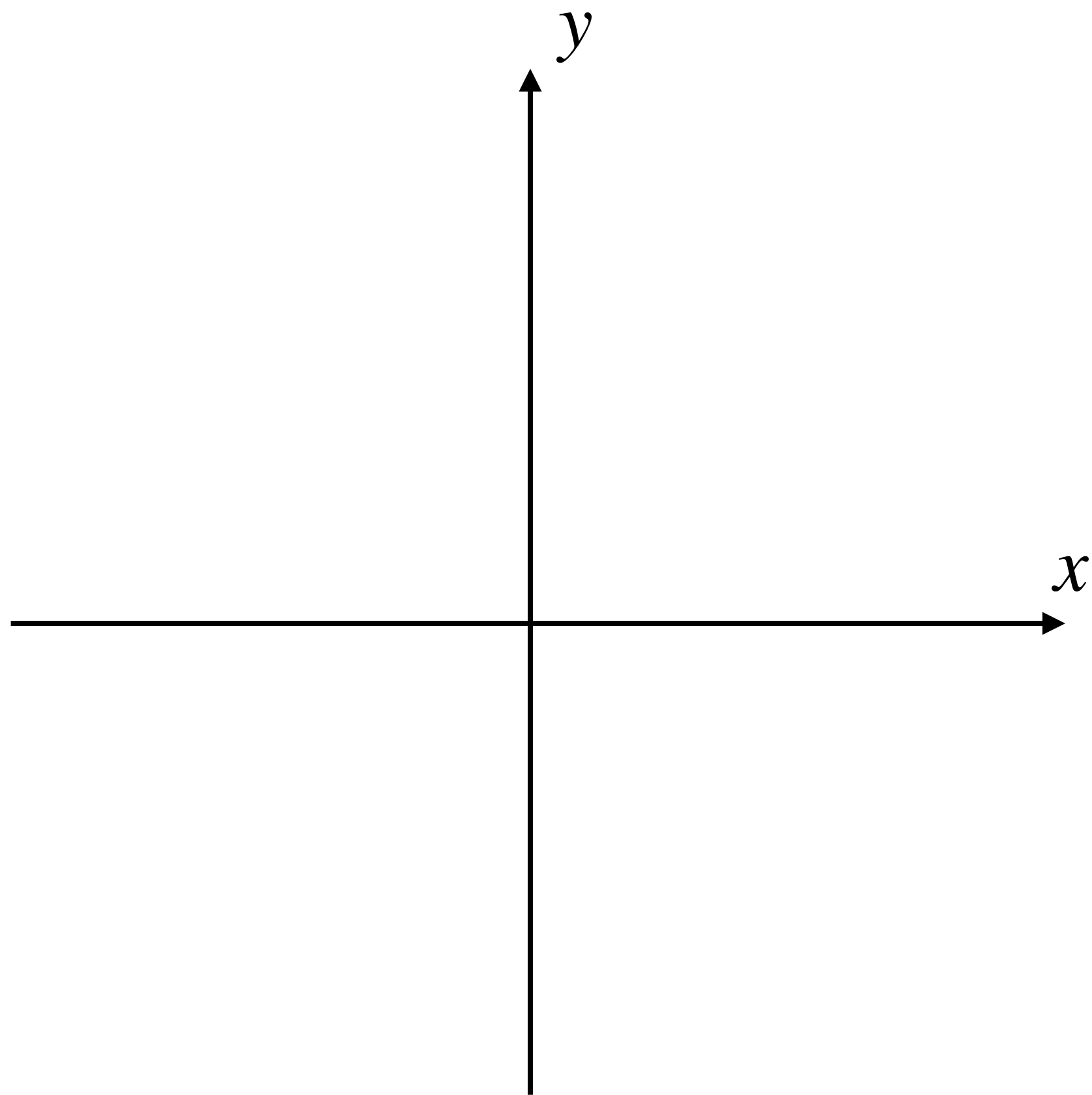
if $0 \leq r \leq 1$, then the transformation pulls points towards the origin

Example: Shearing

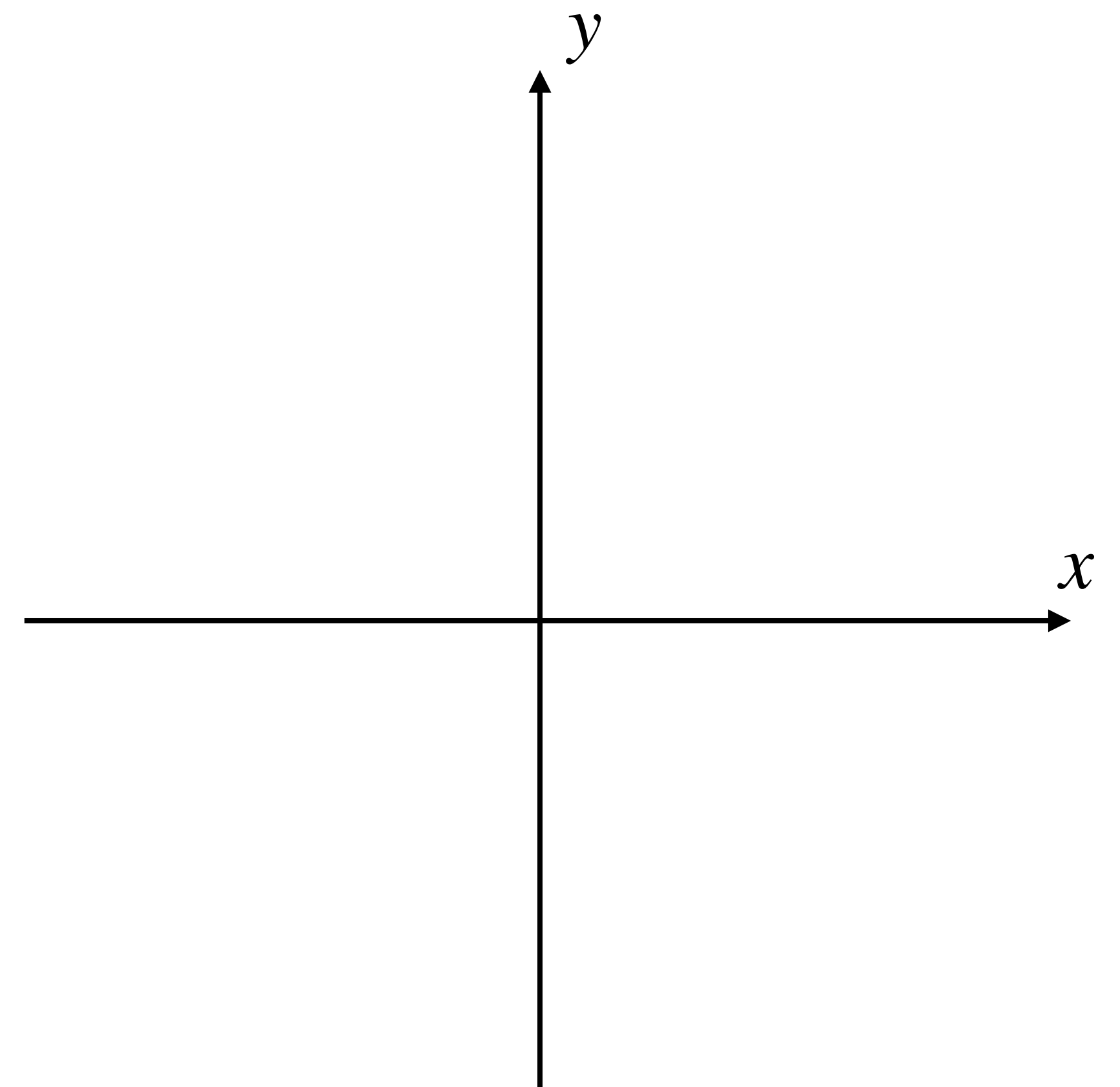


Example: Shearing

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$

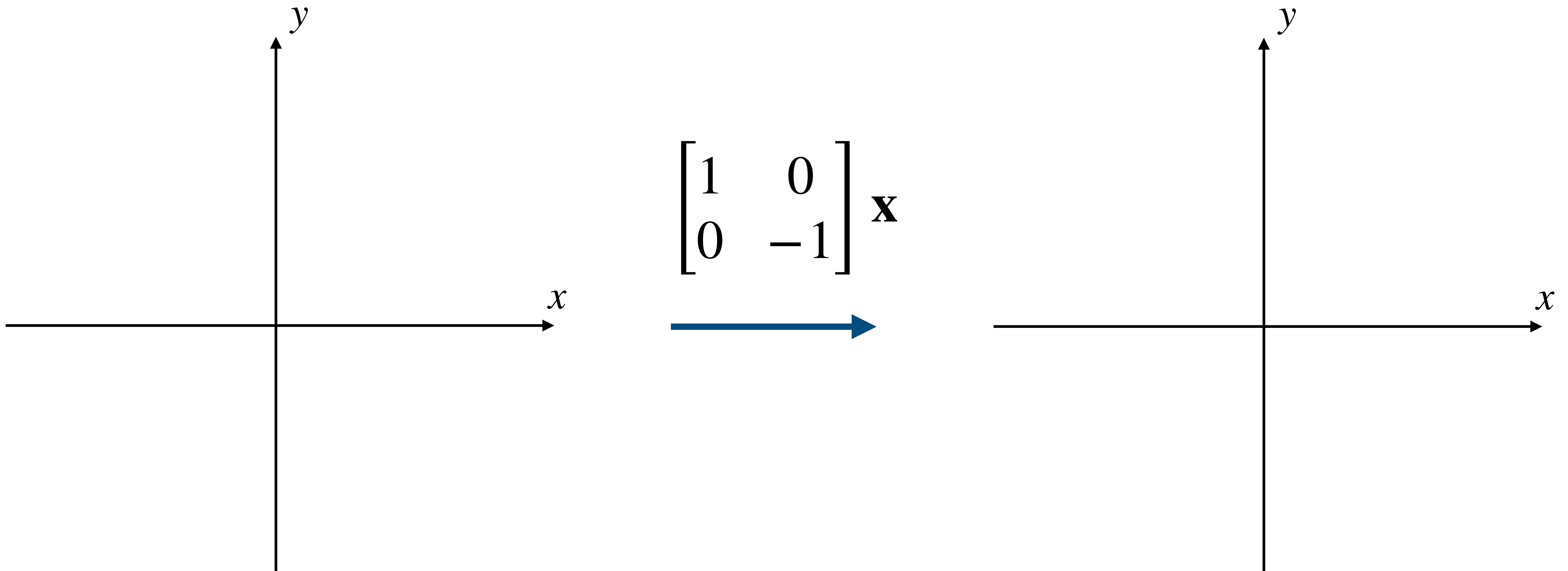



Question

$$\mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$

Draw how this matrix transforms points. What kind of transformation does it represent?

Answer: Reflection



demo

Summary

Matrices can be viewed as **linear transformations**

Matrix transformations change the **shape** of points sets

Linear transformations behave well with respect to **linear combinations**